

Beginning Algebra Final Test (Part I)

Name
Date
Course

Instructions

- i) As always, write a title, name, date, and course.
- ii) Number each problem, circle the numbers, and show work, even when you can do the problems in your head.
- iii) Draw a rectangle around your answers.
- iv) Staple a copy of these test questions with your work and your final answers.

Solve.

$$\textcircled{1} 3x^4 + 3x^3 = 36x^2 \quad \textcircled{2} 5y^3 = 500y$$

$$\textcircled{3} -12x^2 - 9x + 27 = -4x^3 \quad \textcircled{4} 0 = w^4 - 3w^2 - 4$$

Perform the indicated operations and simplify completely.

$$\textcircled{5} \frac{y^2 - 2y}{10x^2 - 49xy + 18y^2} \div \frac{2 - y}{4x^2 - 81y^2}$$

$$\textcircled{6} \frac{-4c + 7}{8c^4 + 8c^3} + \frac{1}{4c^2}$$

$$\textcircled{7} \frac{x+6}{x-2} - \frac{x+4}{x+2} \quad \textcircled{8} \frac{y^5}{3+2y} \cdot \frac{6y^4 - 9y^3}{-22y^3 + 31y^2 + 3y}$$

- 9) a) Simplify the following polynomial.
- b) Write the polynomial in descending order of powers.
- c) State the polynomial's degree.

$$3x^2 + 1 - 4x^3 + 5x^3 + 8 - 4x - 7x^3 + 9$$

$$\text{d) Expand: } (y+7)^2$$

Perform the indicated operations and simplify completely.

$$\textcircled{10} \text{ Divide: } \frac{16x^5 + 40x^4 + 56x^3 - 8x^2}{-8x^2}$$

$$\textcircled{11} \text{ Expand: } (3x-4)(x^2-2x+1)$$

$$\textcircled{12} \text{ a) Subtract: } (-7x^2 + 4x - 7) - (3x^2 + x - 10)$$

$$\text{b) Multiply: } 6x^4(-x^4 + 6x^3 - 7x^2 + 3)$$

$$\textcircled{13} \frac{3}{4c} - \frac{7c}{4c+4} + \frac{-8}{c^2+c} \quad \textcircled{14} \text{ Simplify: } \frac{4x^3 + 4x^2 - x - 1}{4x^2 - 1}$$

Rationalize the denominators.

$$\textcircled{15} \frac{-4}{\sqrt{10}} \quad \textcircled{16} \sqrt{\frac{3}{4}} \quad \textcircled{17} \sqrt{\frac{3}{8}} \quad \textcircled{18} \sqrt{\frac{27}{32}}$$

19) Simplify the expressions.

$$\text{a) } \sqrt{64} \quad \text{b) } 7\sqrt{2}(-3\sqrt{6}) \quad \text{c) } \sqrt{\frac{4}{9}}$$

$$\text{d) } 3\sqrt{56} - 2\sqrt{14} + \sqrt{126}$$

Simplify.

$$\textcircled{20} \frac{-3b^6}{\frac{2}{a^2b}}$$

$$\textcircled{21} \frac{\frac{2x-10}{x^2+4x+3} + \frac{1}{x+3}}{\frac{7}{x+1} + 5}$$

For the problems below, always label the axes of your graphs and the intervals on each axis. Write all coordinates you use to make your graphs (e.g. (2,-3)).

②② Graph the following equation using the slope and y-intercept.

$$y = -\frac{3}{4}x + 2$$

②③ Graph the following equation using a point visible in the equation and the slope.

$$y + 1 = 2(x - 1)$$

②④ Graph the following equation using x and y intercepts.

$$5x + 4y = -5$$

②⑤ Graph the following equation using any method.

$$3x = 2y - 5$$

Beginning Algebra Final Test (Part I) Answers

Name
Date
Course

① $x = -4$, $x = 3$, $x = 0$

② $y = 10$, $y = -10$, $y = 0$

③ $x = \frac{3}{2}$, $x = -\frac{3}{2}$, $x = 3$

④ $w = 2$, $w = -2$

⑤ $\frac{-y(2x+9y)}{5x-2y}$ or $-\frac{y(2x+9y)}{5x-2y}$ or $\frac{y(2x+9y)}{-(5x-2y)}$

⑥ $\frac{2c^2 - 2c + 7}{8c^3(c+1)}$

⑦ $\frac{6x+20}{(x-2)(x+2)}$ or $\frac{6x+20}{x^2-4}$

⑧ $\frac{-3y^7}{(3+2y)(11y+1)}$

⑨ a) & b) $-6x^3 + 3x^2 - 4x + 18$ c) $D = 3$
d) $y^2 + 14y + 49$

⑩ $-2x^3 - 5x^2 - 7x + 1$

⑪ $3x^3 - 10x^2 + 11x - 4$

⑫ a) $-10x^2 + 3x + 3$ b) $-6x^8 + 36x^7 - 42x^6 + 18x^4$

⑬ $\frac{-7c^2 + 3c - 29}{4c(c+1)}$

⑭ $x+1$

⑮ $-\frac{2\sqrt{10}}{5}$

⑯ $\frac{\sqrt{3}}{2}$

⑰ $\frac{\sqrt{6}}{4}$

⑱ $\frac{3\sqrt{6}}{8}$

⑲ a) 8 b) $-42\sqrt{3}$ c) $\frac{2}{3}$ d) $7\sqrt{14}$

⑳ $\frac{-3a^2b^7}{2}$

㉑ $\frac{3x-9}{5x^2+27x+36}$



